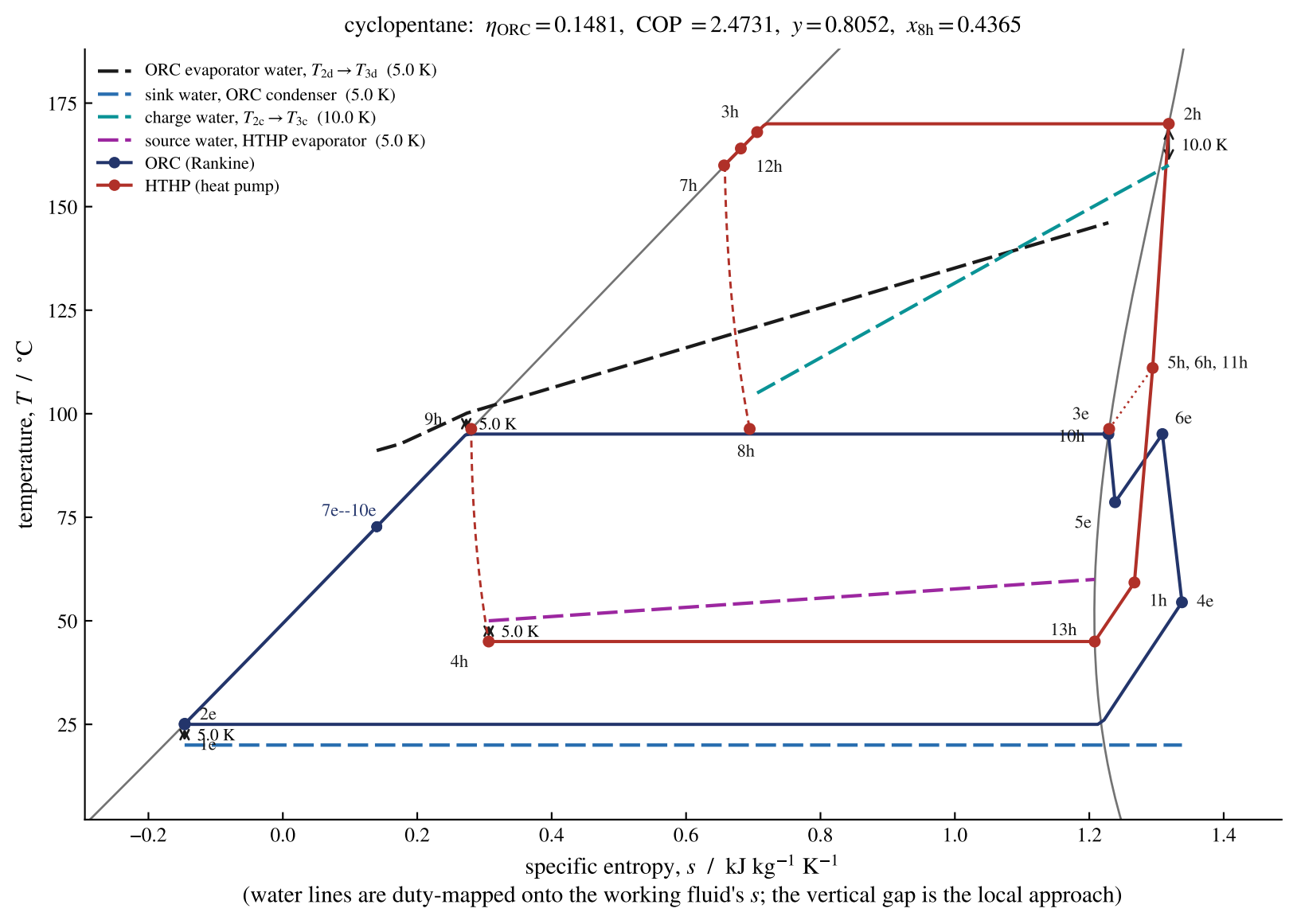
cyclopentane: $\eta_{ORC} = 0.1481$, COP $= 2.4731$, $y = 0.8052$, $x_{8h} = 0.4365$
ORC evaporator water, $T_{2d} \to T_{3d}$ (5.0 K)
sink water, ORC condenser (5.0 K)
charge water, $T_{2c} \to T_{3c}$ (10.0 K)
source water, HTHP evaporator (5.0 K)
ORC (Rankine)
HTHP (heat pump)
temperature, T / °C
specific entropy, s / kJ kg^{-1} K^{-1}
(water lines are duty-mapped onto the working fluid's s; the vertical gap is the local approach)
3h
2h
10.0 K
12h
7h
5h, 6h, 11h
9h
5.0 K
3e
6e
10h
8h
7e--10e
5e
1h
4e
13h
5.0 K
4h
2e
5.0 K
1e